

Stylized Facts about AI Capabilities

Tom Cunningham, METR Ali Merali, Yale

Background

My background. IIES Stockholm → Facebook → Twitter → Integrity Institute → OpenAI Economic Research → METR.

tom.cunningham@gmail.com

My position.

1. If AI becomes as smart as humans it will be enormously disruptive.
2. It's plausible that this will happen in the next decade.

Founded in 2022. Founded by ex-OpenAI researchers, building evals & understanding on how to detect worrying AI capabilities, especially ability for AI to do AI research. We're hiring!

Model Evaluation and Threat Research (METR)

Time horizon metric.

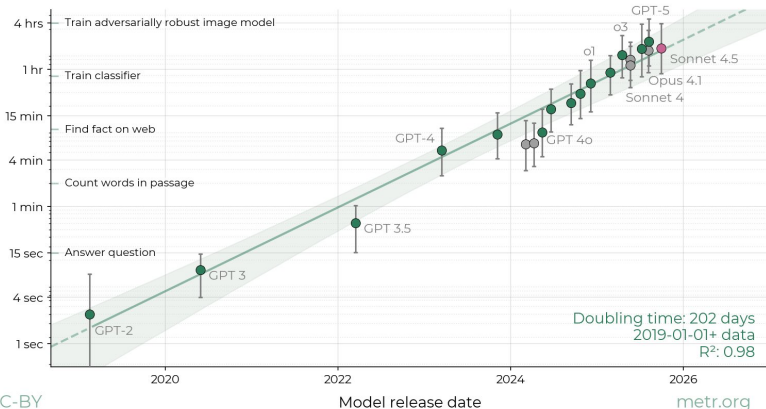
Idea: quantify LLM capability by sorting tasks according to “time horizon” (how long an expert human takes to do the task).

For each model find the horizon-length at which the LLM has a 50% success rate.

Claude Sonnet 4.5 has a 50%-time-horizon of about 1 hr 53 min (95% CI: 50 - 235 min)
Task length (at 50% success rate)



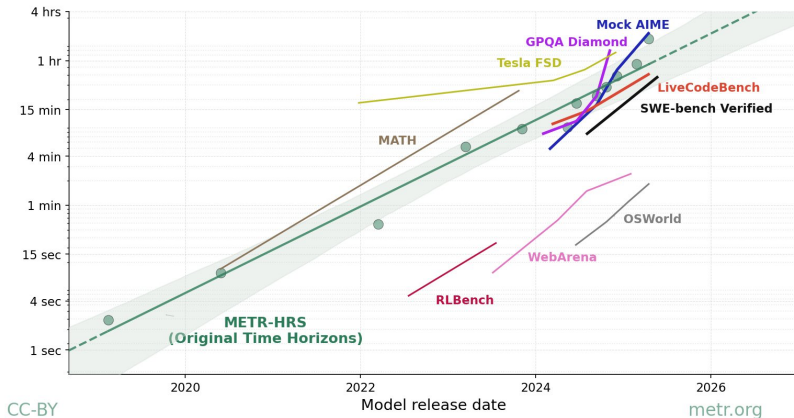
METR



Model Evaluation and Threat Research (METR)

AI time horizons are increasing in many domains

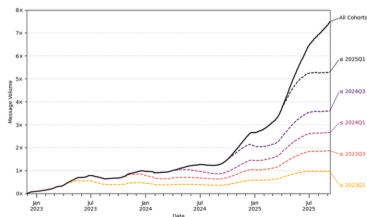
Task length (at 50% success rate)



How People Use ChatGPT

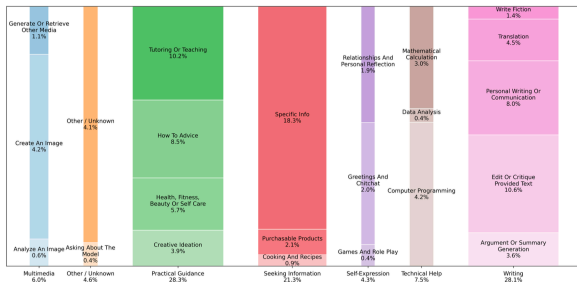
800M weekly ChatGPT users.

Growth in both extensive and intensive margins despite substantial increase in competition.



Around 2/3 of messages are not work-related.

GenAI's primary impact is *home production*, solving everyday problems. Plausible that it's a substitute for market labor.



Stylized Facts about AI Capabilities

Genesis of this work.

In summer 2024 OpenAI was preparing to launch the first reasoning model (strawberry/o1-preview), I was asked to help estimate the likely economic impact.

The new model performed far better on reasoning-intensive tasks (MATH, GPQA), comparable to a PhD student. What will be the impact of a model with PhD-level intelligence?

Ideal: plug a vector of capabilities into an economic model, & predict change in equilibrium.

Stylized Facts and Implications

Fact about AI	Implication for Economics of AI
1. We have no standard metric for computer intelligence	Opportunity for theorists.
2. LLM capabilities are highly correlated.	Directed technical change may be difficult.
3. LLM capabilities are growing rapidly.	Diffusion isn't primary cause of growth.
4. LLM capabilities are distinct from human capabilities.	Need a two-factor model of ability.
5. There is no consensus on the bottleneck.	Opportunity for theorists.
6. Inference prices are falling at 10X/year.	GDP share of compute may be small.
7. LLMs can either do a task cheaply or not at all.	Unlikely to be equilibrium task-sharing

We have no standard metric for computer intelligence.

A graveyard of measures of human intelligence:

Measure	Interpretation	Year Passed
Chess	Shannon: <i>“Chess is generally considered to require ‘thinking’ for skilful play.”</i>	1997
Winograd	<i>“when people pass the test, we would say they were thinking.”</i>	2019
ARC-AGI	<i>“the only formal benchmark of AGI progress.”</i>	2024
MMMU	<i>“Benchmark for Expert AGI.”</i>	2025
AGIEval	<i>“general abilities ... in tasks pertinent to human cognition”</i>	2025

There are some live proposals, none are canonical.

- Morris et al. (2023) (Google DeepMind) five “levels of AGI.”
- OpenAI five “levels of AGI,” not officially released.
- Hendrycks et al. (2025) “the definition of AGI.”

Most definitions are either overly narrow or ambiguous.

This is a hard problem!

It’s intellectual quicksand, many smart people have got themselves stuck. Sensible to tread carefully.

Available Metrics of AI capability

Two common metrics of AI capability.

1. **Scores on benchmarks.** There are hundreds of different benchmarks, however no canonical set, and they often get “saturated” relatively quickly. Continual scramble to create new benchmarks.
2. **User pairwise preference.** Given two answers to a query, which do you prefer? These are used by labs to measure user utility, also some 3rd-party data sources, esp. LMs chatbot arena.

Other metrics of AI capability.

1. **Causal effect on user retention.** Used by labs but the results are not generally reported.
2. **Uplift effects on human productivity.** Hard to compare over time because you need the same humans.
3. **User willingness to pay.** Few studies, Brynjolfsson reports \$100/user/month.
4. **Adoption.** Hard to infer value.

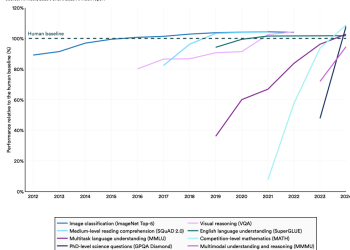
LLM capabilities are growing rapidly

Benchmarks are falling rapidly.

It typically takes 18 months for LLM scores on a benchmark to go from 25% to 75%.

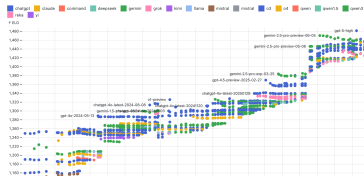
Select AI Index technical performance benchmarks vs. human performance

Source: AI Index, 2023 | Chart: 2023 AI Index report



User preference scores are growing rapidly.

Models have 70% win-rate over 1yo models (150 Elo/year).



Implication for Economic Forecasts

There are many recent forecasts of AI's economic impact.

Economists typically forecast 0-1.5% range, AI people typically 3-30%+.

date of forecast	author	annual excess growth, 2025-2035
May 2023	Bailey, Brynjolfsson, and Korinek (2023)	+2.8%/year
June 2023	McKinsey Global Institute (2023)	+0.1-0.6 pp/yr ¹
August 2023	Tyler Cowen	+0.25%-0.5% ²
2023	Briggs & Kodnani / Goldman Sachs ³	+1.5%/year
March 2024	Commission de l'Intelligence Artificielle (2024)	+1.3 pp/yr ⁴
March 2024	Korinek and Suh (2024)	+10%/year
April 2024	Acemoglu (2024)	+0.07%/year
May 2024	Aldasoro et al. (2024)	+1.0-1.5 pp/yr ⁵
June 2024	Aghion and Bunel (2024)	+1%/year
Dec 2024	Filippucci, Gal, and Schief (2024)	+0.25-0.6 pp/yr TFP ⁶
Feb 2025	Tyler Cowen	+0.5% ⁷
March 2025	Office for Budget Responsibility (2025)	+0.3 pp/yr ⁸
March 2025	Bergeaud et al. (2025)	+0.35 pp/yr ⁹
March 2025	Erdil et al. (2025)	+30%/year ¹⁰
April 2025	Misch and Zhang (2025)	+1.1% ¹¹
May 2025	Jack Clark	+3-5% ¹²
June 2025	Filippucci et al. (2025)	+0.4-1.3 pp/yr ¹³
June 2025	OECD (2025)	+0.2-1.3 pp/yr ¹⁴
Sept 2025	Penn Wharton Budget Model	+0.15% ¹⁵
Oct 2025	Andrej Karpathy	no change in trend ¹⁶

Implication for Economic Forecasts

Economists' forecasts typically assume AI is a *one time event*.

Various assumptions about speedups across tasks (from Trammell (2025)):

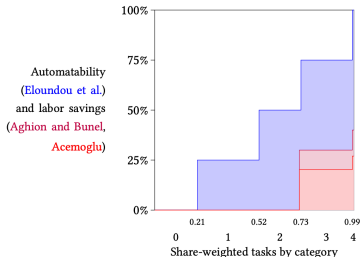


Figure 1: Distribution of “automatability” (Eloundou et al.) and “labor cost savings” (Acemoglu, Aghion and Bunel) across share-weighted tasks

Acemoglu (2024) and Aghion and Bunel (2024) derive growth rates by assuming diffusion speeds.

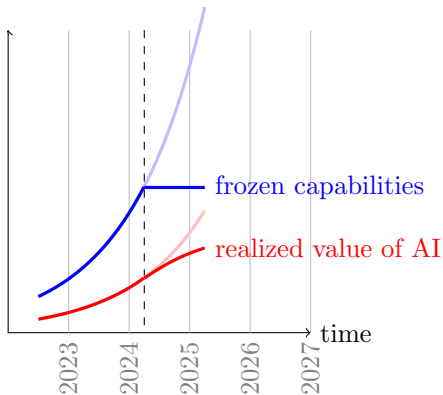
But AI capabilities are rapidly increasing – these areas are expanding.

Implication for Economic Forecasts

Much growth is capabilities, not diffusion.

Thought experiment: if we had frozen capabilities at gpt4o (May 2024), how much smaller would the impact be today?

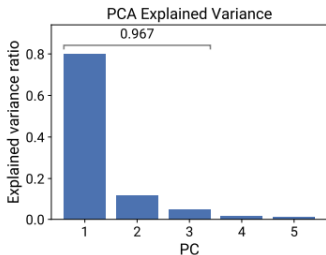
Some evidence from AB tests on platforms, which compare engagement with different models.



LLM capabilities are highly correlated

Many studies find high correlations across scores.

E.g. Ruan, Maddison, and Hashimoto (2024) finds the first factor explains 80% of variance:



(a) PCA explained variance

There is also high correlation between scores on benchmarks and scores on user-preference rankings.

LLM capabilities are highly correlated

What does capability correlation mean concretely?

The model that is best at math is also probably best at programming, medical advice, legal advice, general knowledge, etc.

It was not obvious that this would have happened. The best program at chess is not the best program at jeopardy, and vice-versa.

Why should this be so?

A common interpretation is based on bitter lesson / manifold hypothesis / scaling laws: training a larger model makes it better at everything, by recovering deep latent structure of the world.

Qualifications.

1. There is still a tradeoff between *cost* and ability (usually cost and speed are highly correlated).
2. Different frontier models do have different specialities. However typically any frontier model today will beat *all* frontier models from 6 months ago.
3. Fine-tuning on proprietary data can make capabilities spiky.
4. Post-training with RL may induce less correlation.

Implication: we may not be able to direct technical change

Should we direct AI towards augmenting instead of automating?

Brynjolfsson (2023): *“an excessive focus on developing and deploying [human-level AI] can lead us into a trap ... when AI is focused on augmenting humans rather than mimicking them, then humans retain the power to insist on a share of the value created”*

Acemoglu (2021): *“our current trajectory automates work to an excessive degree while refusing to invest in human productivity.”*

Our ability to direct change may be limited.

If there is truly a single latent factor of model ability, then work to train an augmenting model looks very similar to work to train an automating model, & the ability to steer AI capabilities may be limited.

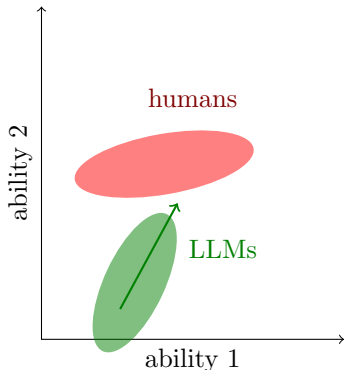
LLM capabilities are distinct from human capabilities

Ideally we would compare human and LLM performance on the same tasks.

If we had a full matrix of human and LLM performance we would be able to compare latent factors of each. Unfortunately there are few datasets which collect human performance across a range of distinct benchmarks.

Qualitatively, LLM and human abilities appear very different.

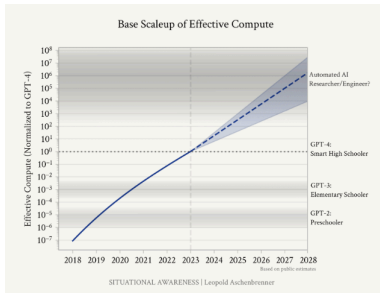
Aptitude tests that are predictive for humans are not predictive for LLMs. GPT-4 was able to get human-level scores on SAT, LSAT, civil service exams, but not able to do those jobs.



Implication: a single-factor model will be a poor fit

We sometimes describe AI capability by human equivalent.

E.g. years of education (Aschenbrenner (2024))



Some models assume a common human/AI skill factor.

E.g. Ide and Talamas (2024).

What would a two-factor model look like?

- LLMs have PhD-level knowledge and reasoning.
- LLMs have preschool-level tactics.

There is no consensus on the bottleneck

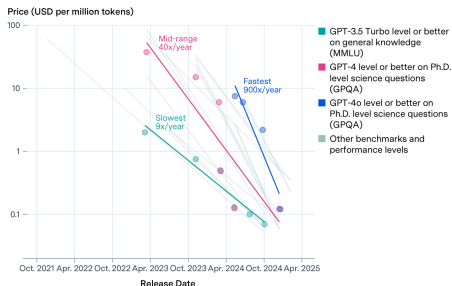
Bottleneck	Status
Cost.	Does not seem an important bottleneck.
Deductive reasoning.	Much improved since 2024.
Information-gathering	Much improved since 2024.
Accuracy / hallucination.	Much improved since 2024.
Long context.	Much improved since 2024.
Multimodal inputs and outputs.	Much improved since 2024.
In-context learning.	Much improved since 2024 (ARC-AGI).
Continual learning.	Amodei, Karpathy, Hassabis.
Novel situations	Vafa et al. (2024) on world models.
Interactive decision-making.	Still struggles.
Human predictability	Jagged frontier, Vafa et al. (2024)

Note: once a problem has been well-defined, LLMs have a very strong track record of solving it.

Inference prices are falling at 10X/year

From Artificial Analysis.

For given capabilities inference prices have been falling at 10X+/year.



Why are they falling?

Not because of lower margins – we see the same trend looking at raw compute cost for open models.

Smaller models are getting better through (1) better training techniques; (2) more training compute; (3) more training data; (4) better training data.

LLMs can either do a task cheaply or not at all

Ability to do tasks.

METR (2024) say “when agents *can* do a task, they can often do so at a fraction of the cost of humans.”

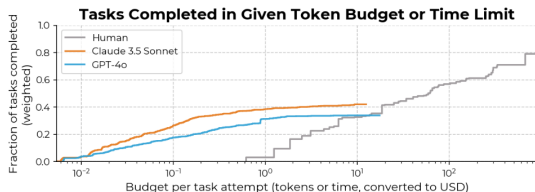


Figure 6: When agents can do a task, they can often do so at a fraction of the cost of humans. The diminishing marginal improvements in agent performance as a function of cost are evident here, contrasting with the log-linear relationship observed in our human baseliners.

Some exceptions to this rule.

1. Voice transcription for a brief period.
2. Some reasoning tasks (ARC-AGI), briefly.
3. Fixed costs (Svanberg et al. (2024)).

Implication: wages are unlikely to be pinned down by the price of compute

**Some models
feature
equilibrium
task-sharing:**

Humans perform tasks that computers are able to do, but it is too expensive for computers:

- Trammell and Korinek (2023)
- Ide and Talamas (2024)
- Smith (2024)
- Acemoglu (2024)

**Equilibrium
task-sharing would
require either:**

1. A dramatic increase in AI prices to perform a task:
 - An increase in the compute required for new tasks (the price of frontier inference has been steady).
 - An increase in compute costs (GPU rental prices have been steady, and the purchase/rent ratio has been stable).
 - An increase in margins.
2. A dramatic drop in human wages.

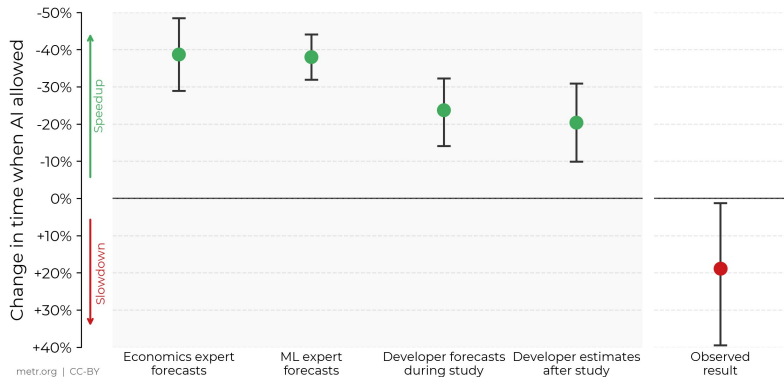
End

METR Uplift Study

Against Expert Forecasts and Developer Self-Reports, Early-2025 AI Slows Down Experienced Open-Source Developers



In this RCT, 16 developers with moderate AI experience complete 246 tasks in large and complex projects on which they have an average of 5 years of prior experience.



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